

$$f: M \rightarrow \mathbb{R}^3: \text{frontal.}$$

$$\Leftrightarrow (f, \nu): M \rightarrow \mathbb{R}^3 \times S^2$$

$$\mathbb{R}^3 \times S^2$$

Legendre map.

$$\theta: \text{canonical contact form.}$$

$$df(v) \cdot \nu = 0 \quad (v \in TM)$$

$$\parallel$$

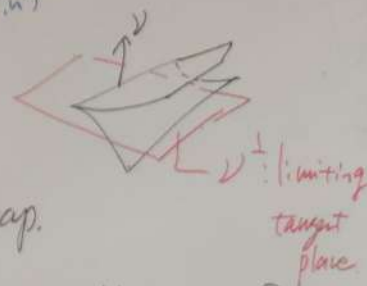
$$(L^* \theta)(v)$$

$$\theta = dx^1 \wedge n^1 + dx^2 \wedge n^2 + dx^3 \wedge n^3$$

(x^1, x^2, x^3) (n^1, n^2, n^3)

$$T\mathbb{R}^3 = \mathbb{R}^3 \times S^2$$

contact mfd



Def. $Y^{2m+1}: \frac{\mathbb{R}^3 \times \mathbb{R}^2 \times \pi}{\sim} \text{mfd.}$

$$Y^{2m+1}: \text{contact mfd.} \Leftrightarrow \exists \theta: 1\text{-form. s.t. } \theta \wedge \left(\bigwedge^m d\theta \right) \neq 0.$$

$(2m+1)\text{-form.}$

Def. $L: \Sigma^m \rightarrow Y^{2m+1}$ is a Legendre map

$$\Leftrightarrow L^* \theta = 0. \quad (\Leftrightarrow \theta(dL(v)) = 0)$$

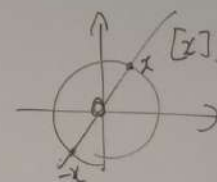
$(v \in T\Sigma^m)$

N^{m+1} : $(m+1)$ -mfd.

T^*N^{m+1} : cotangent bundle. $(2m+2)$ -dim.

$$\{(p, \alpha) \mid p \in N, \alpha \in (T_p N)^*\}$$

$$L = (f, v)$$



$$\alpha : T_p N \xrightarrow{\text{linear}} \mathbb{R}$$

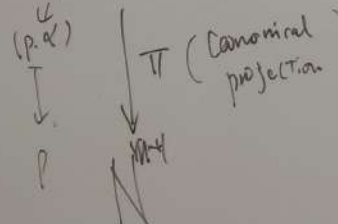
$\mathcal{P}(T^*N^{m+1})$: T^*N^{m+1} の projectivization 射影化.
 $(2m+1)$ -dim. $\{[x] \mid x \in T^*N^{m+1} \setminus \{0\}\}$
 これは "canonical contact form" θ がある.

$$x \sim x' \iff \exists \lambda \in \mathbb{R} \setminus \{0\} \text{ s.t. } x' = \lambda x.$$

Def. A map $f : \Sigma^m \rightarrow N^{m+1}$ is a frontal (front)

$$\iff \exists L : \Sigma^m \xrightarrow{c^a} \mathcal{P}(T^*N^{m+1}) \begin{cases} \text{immersion} \\ \text{s.t.} \end{cases} \begin{cases} L^* \theta = 0. \text{ (Legendre condition)} \\ \pi \circ L = f. \text{ (lift condition)} \end{cases}$$

★ frontal / front は、
 外空間 N^{m+1} の
 可微分構造を定めておく。



Fukuma
帰論

Claim

退化しない n 特異点は
定義域と値域の座標変換
 に依存しない概念である

$$f, \tilde{f}: U \rightarrow \mathbb{R}^3$$

$$S(f) = \{v = 0\}$$

$$\lambda = \det(f_u, f_v, \nu)$$

$$\tilde{\lambda} = \det(\tilde{f}_u, \tilde{f}_v, \tilde{\nu})$$

$$(\tilde{f} := \Phi \circ f)$$

$$(*) : \tilde{\lambda} = (\text{non-zero}) \cdot \lambda$$

Proof of claim

$$\lambda = (\text{non-zero}) \cdot \nu^n \text{ かつ } \nu \in \mathbb{R}$$

$$(*) \text{ 8) } \tilde{\lambda} = (\text{non-zero}) \cdot \tilde{\nu}^n \text{ かつ } \tilde{\nu} \in \mathbb{R}$$

$\tilde{f}$ は退化しない n 点

(*) の proof

$$f: U \rightarrow \mathbb{R}^3 : \text{smooth}$$

$$\Phi: \mathbb{R}^3 \rightarrow \mathbb{R}^3 : \text{diff}$$

$$\tilde{f} := \Phi \circ f$$

$\tilde{f}$ の unit normal $\tilde{\nu}$ は

$$\tilde{\nu} = \frac{(J^T)^{-1} \nu}{\|(J^T)^{-1} \nu\|}$$

$$J^T = (A_x) \times (A_y) =$$

(*) の proof

$$f: U \rightarrow \mathbb{R}^3 : \text{smooth}$$

$$\Phi: \mathbb{R}^3 \rightarrow \mathbb{R}^3 : \text{diff.}$$

$$\tilde{f} := \Phi \circ f = \tilde{f}$$

$\tilde{f}$ の unit normal $\tilde{\nu}$ は

$$\tilde{\nu} = \frac{(J^T)^T \nu}{\|(J^T)^T \nu\|} \quad (J = d\Phi_f)$$

$$J^T = (A_x) \times (A_y) = (\det A) (A^T)^T (x \times y)$$

$$\tilde{\lambda} = \det(\tilde{f}_u, \tilde{f}_v, \tilde{\nu})$$

$$= \frac{1}{\|(J^T)^T \nu\|} (\tilde{f}_u \times \tilde{f}_v) \cdot (J^T)^T \nu$$

$$= \frac{1}{\|(J^T)^T \nu\|} ((J f_u) \times (J f_v)) \cdot (J^T)^T \nu$$

$$= \frac{\det J}{\|(J^T)^T \nu\|} ((J^T)^{-1} (f_u \times f_v)) \cdot (J^T)^T \nu$$

$$f_u \times f_v = \lambda \nu$$

$$\begin{aligned} &= \frac{\det J}{\|0\|} \frac{((J^T)^{-1} \lambda \nu) \cdot (J^T)^T \nu}{\|(J^T)^T \nu\|^2} \\ &= \frac{\det J}{\|(J^T)^T \nu\|} \lambda \frac{\|(J^T)^T \nu\|^2}{\|(J^T)^T \nu\|^2} \\ &= (\det J) \cdot \|(J^T)^T \nu\| \cdot \lambda \end{aligned}$$

↑ (non-zero)
 $\text{by } (*) \text{ and } \tilde{\nu}$

Intrinsic invariants of (3,4)-cuspidal edge.

- 平行曲面上に現れる重要な特異点だが、
- 特異曲率は符号が well-defined である。
- とくに凸凹を測る内在的不変量を定める必要あり。



- K_s, β, δ : 内在的不変量
 ↑ strongly normalized adapted coordinate system.
 (snacs) の導入: $5/26$ 入

to do snacs の導入は K_s, β, δ の coord を β, δ 書く。

- K の有界性 $\iff \beta = \delta = 0$ $6/23$ 入
- β の正負と (3,4)-CE の凸凹 (MST の normal form) $5/4$ 入

≡ 2.7.10

- (3,4)-CE, (3,5)-CE の内在的判定条件

$\checkmark$ (3,4)-CE $\iff$ 退化記号2 第一種, $W_{34} \neq 0$
 3-type edge front

$\checkmark$ (3,5)-CE $\iff$ 3-type edge, front 2/3, $(W_{34} = 0)$

$\Pi_{3,4} = -\frac{1}{2}\beta$
 β is intrinsic

$\Pi_{3,5} = -\frac{1}{2}\delta - 2L_3W_{3,4} - \frac{G_{3,5}}{24}\beta$
 $\Pi_{3,5}$ is intrinsic かも
if $\beta=0, K \neq 0$ then $\Pi_{3,5} = -\frac{1}{2}\delta$
intrinsic: OK かも

product curvature of 3 and 5

- $\checkmark$ generic (3,4)-CE $\iff$ 退化記号2 第一種, $W_{34} \neq 0$
- $\checkmark$ generic (3,5)-CE $\iff$ " " $\Pi_{3,4} = 0$
 $\Pi_{3,5} \neq 0$

⊗ の proof

Cor

f, g : generic frontal, isometric

Remark

(4.5)-CE には
このような主張
は成り立たない

[HHM]

Cor 4.9

だから 3-type edge

で成り立つこと

証明するべき

$$(1) f: (3.4)\text{-CE} \iff g: (3.5)\text{-CE}$$

⌚ f は 退化二次形式, 第一種, $\underbrace{\Pi_{3,4}^f \neq 0}_{\text{intrinsic}}$ をみたす

$$EG - F^2 = \lambda^2$$

$$\lambda = (\text{non-zero}) v^2$$

$$\eta = \partial/\partial v$$

$\iff g$ も 退化二次形式, 第一種, $\Pi_{3,4}^g \neq 0$

$$(2) f: (3.5)\text{-}$$

$$f \text{ pr } (3.5)\text{-CE}$$

$$\implies \underline{f \text{ is}}$$

$$\implies \underline{g \text{ is}}$$

$$\implies \underline{g \text{ pr}}$$

Isometric

$f: (3,5) - CE$

第一種, $\pi_{3,4}^f \neq 0$ $\Sigma \pi = 0$
intrinsic

$(10) \checkmark$

第一種, $\pi_{3,4}^g \neq 0$

$$(2) f: (3,5) - CE \iff g: (3,5) - CE$$

$f \text{ pr } (3,5) - CE$

$\implies$ f is also not 2, 第一種, $\pi_{3,4}^f \equiv 0$, $\pi_{3,5}^f \neq 0$ $\Sigma \pi = 0$

$\implies$ g is " ", $\pi_{3,4}^g \equiv 0$, $\pi_{3,5}^g \neq 0$

$\implies g \text{ pr } (3,5) - CE //$

HHMの計算
Σλ=3 (9/41+)

$\omega_{m,m+1}$

Rem

$\pi_{3,5}$

た

T

V

R

R

$\boxed{\begin{matrix} 5/20 \\ 1-1 \end{matrix}}$

(3,5) - CE

HHMの計算
Σλ_{3,5} (1/4 1+)

$$\omega_{m,m+1} \text{ 対 } \omega_{m,m+1} = \dots = \omega_{m,m+1} = 0$$

27502 well-def.

← appendix 2" 示す

種重, $\pi_{3,4}^f \equiv 0, \pi_{3,5}^f \neq 0$ Σ27502

↓ OK

$\pi_{3,4}^g \equiv 0, \pi_{3,5}^g \neq 0$

$\frac{5}{20}$
1-1

Rem

$\pi_{3,5}$ は extrinsic inv

だpl. $K_D \neq 0, \pi_{3,4} \equiv 0$ 対 5

$\pi_{3,5} = -\frac{1}{2} \delta$ と対

内在的判定に便える

残っている case

$K_D \equiv 0 (\Rightarrow \beta \equiv 0)$

$\omega_{3,4} \neq 0$ のとき

$\pi_{3,5}$ は intrinsic?

Problem 22 残った

Rem

$\pi_{3,4} \equiv \pi_{3,5} \equiv 0$ だpl

K 非有界の例あり

★ この時は $K_D \equiv 0$ generic 27502